6.S974 Games, Learning, and Security
HW2

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Problem 1 (Cybersecurity with Incomplete Information)

(a) **Pure strategies.** A strategy specifies a choice for each type.

- Attacker ($\theta_1 = 4, \theta_1 = 8$): $\{(A, A), (A, N), (N, A), (N, N)\}$
- Company: ($\theta_2 = 1, \theta_2 = 3$): $\{(P, P), (P, U), (U, P), (U, U)\}$

A strategy profile is a tuple of one pure strategy per player. In this case each player has four pure strategies, so there are $4 \times 4 = 16$ strategy profiles.

(b) **Prior and beliefs.** Types are equally likely to occur and independent so

$$p(\theta_1, \theta_2) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4}, \qquad \forall\, \theta_1,\ \forall\, \theta_2$$

Beliefs are conditional on the attacker or the company's types

$$\mu_1(\theta_2 = 1 \mid \theta_1 = 4) = \frac{p(4,1)}{p(4,1) + p(4,3)} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \tfrac{1}{2}$$

Because the prior is uniform over (θ_1, θ_2), this belief holds for every player's type for every other player's type so

$$\mu_1(\theta_2 \mid \theta_1) = \mu_2(\theta_1 \mid \theta_2) = \tfrac{1}{2}$$

(c) **Pure strategy best response.** Expected payoff when company plays P is $E_{A\sim\alpha}[u_2(P, A)] = -\alpha\theta_2 - (1-\alpha)\theta_2 = -\theta_2$ and when company plays U is $E_{A\sim\alpha}[u_2(U, A)] = -4\alpha$, so

$$BR_2(\alpha) = \begin{cases} P & \theta_2 < 4\alpha \\ \{P, U\} & \theta_2 = 4\alpha \\ U & \text{else} \end{cases} \implies \begin{cases} (P,P) & \alpha \geq \frac{3}{4} \\ (P,U) & \frac{1}{4} \leq \alpha \leq \frac{3}{4} \\ (U,P) & \text{never} \\ (U,U) & \alpha \leq \frac{1}{4} \end{cases}$$

And for the attacker we get $E_{P\sim\pi}[u_1(P, A)] = -\pi + (1-\pi)(\theta_1 - 1) = (1-\pi)\theta_1 - 1$ and $E_{P\sim\pi}[u_1(P, N)] = 0$

$$BR_1(\pi) = \begin{cases} A & \pi < 1 - \frac{1}{\theta_1} \\ \{A, N\} & \pi = 1 - \frac{1}{\theta_1} \\ N & \text{else} \end{cases} \implies \begin{cases} (A,A) & \pi \leq \frac{3}{4} \\ (A,N) & \text{never} \\ (N,A) & \frac{3}{4} \leq \pi \leq \frac{7}{8} \\ (N,N) & \pi \geq \frac{7}{8} \end{cases}$$

(d) **Does a pure strategy Bayes Nash Equilibrium exist?**

- Begin with attacker pure strategy (A, A): $\sigma_1(A \mid \theta_1 = 4) = 1 = \sigma_1(A \mid \theta_1 = 8)$ so $\alpha = 1$. From part (c) $BR_2(1) = (P, P)$. By the same logic company pure strategy (P, P) yields $\pi = 1$ $BR_1(1) = (N, N)$. Repeating, attacker pure strategy (N, N) yields $\alpha = 0$ and $BR_2(0) = (U, U)$. Company pure strategy (U, U) yields $\pi = 0$ and $BR_1(0) = (A, A)$, returning us to the start of the cycle. So there's no pure strategy Bayes Nash Equilibrium with any combination of $((A, A), (N, N))$ and $((P, P), (U, U))$
- Take attacker pure strategy (A, N) yielding $\alpha = \frac{1}{2}$ and $BR_2(\frac{1}{2}) = (P, U)$. The company's best response yields $\pi = \frac{1}{2}$ and $BR_1(\frac{1}{2}) = (A, A)$, returning us to the first cycle.
- Take attacker strategy (N, A) yielding $\alpha = \frac{1}{2}$. This produces the same $BR_2(\frac{1}{2}) = (P, U)$ above returning us to the first cycle.

Hence by process of elimination, there is no pure strategy Bayes Nash Equilibrium.

(e) **Bayesian Nash Equilibrium in behavioral strategies.**

Necessary condition $\pi = \frac{3}{4}$ By contradiction suppose $\pi \neq \frac{3}{4}$ in equilibrium. Recall that a mixed action is a best response only if every action in its support is a pure best response by theorem.

Case 1 $(\pi > 3/4)$**.** For any $\pi > 3/4$ attacker type $\theta_1 = 4$ has only a pure best response N (because mixed best response requires pure best response in its support) and $\sigma_1(A \mid \theta_1 = 4) = 0$. So $\alpha = \sigma_1(A \mid \theta_1 = 8)/2$, which implies $\alpha \leq 1/2$.

For any $\alpha \leq 1/2$ the company type $\theta_2 = 3$ has only a pure best response U and $\sigma_2(P \mid \theta_2 = 3) = 0$. So $\pi = \sigma_2(P \mid \theta_2 = 1)/2$, which implies $\pi \leq 1/2$, a contradiction to $\pi > 3/4$.

Case 2 $(\pi < 3/4)$ For any $\pi < 3/4$ the attacker has only a pure best response (A, A) and $\alpha = 1$. For $\alpha = 1$ the company has only a pure best response (P, P) and $\pi = 1$, contradicting $\pi < 3/4$.

Therefore if there exists a Bayesian Nash Equilibrium a necessary condition would be $\pi = 3/4$.

Mixed strategy. The probability of protecting $\pi = 3/4$ gives $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) = 3/2$, which implies at least one company type randomizes, since pure company strategies give $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) \in \{0, 1, 2\}$.

Moreover, for $\pi = 3/4$ the best response of attacker type $\theta_1 = 8$ is A, implying a probability of attack $\alpha \geq 1/2$.

For the company to have a mixed strategy and $\alpha \geq 1/2$, it must be that $\alpha = 3/4$ so that the mixed strategy best response can have a support greater than 1: the company type $\theta_1 = 3$ is indifferent at $\alpha = 3/4$ and type $\theta_1 = 1$ is not.

This support tells us that company type $\theta_2 = 1$ has a pure strategy best response of P and $\sigma_2(P \mid 1) = 1$. Plugging this into our equation for π above we get $\sigma_2(P \mid 3) = 1/2$.

Note that $\sigma_1(A \mid 8) = 1$ follows directly from attacker type $\theta_1 = 8$'s best response of A. Solving $\alpha = 3/4 = (1 + \sigma_1(A \mid 4))/2$ gives $\sigma_1(A \mid 4) = 1/2$.

Therefore if there is a mixed strategy Bayesian Nash Equilibrium it must be that

$$\sigma_2(P \mid 3) = 1/2 = \sigma_1(A \mid 4), \quad \sigma_2(P \mid 1) = 1 = \sigma_1(A \mid 8).$$

Suffice to validate.

Validation. From definition, the strategy profile is a Bayes Nash Equilibrium if for every player and every type the strategy is a best response. At the candidate profile $\pi = 3/4$ and $\alpha = 3/4$.

$$\begin{aligned}
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 4)) = \{A, N\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 4) = \{A, N\} \\
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 8)) = \{A\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 8) = \{A\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 1)) = \{P\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 1) = \{P\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 3)) = \{P, U\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 3) = \{P, U\}
\end{aligned}$$

Each condition above was necessary for equilibrium, so it is the unique BNE.

Problem 2 (Signaling through Safety Certification)

(a) **Game tree.**

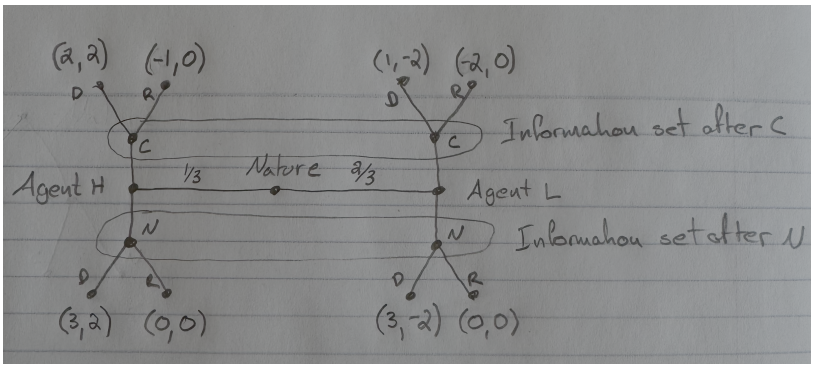


Figure 1: Problem 2 Part (a) game tree

(b) **Does a separating PBE exist?** A separating PBE exists if a strategy profile (s^*, μ^*) satisfies

1. *Sequential rationality.* At every information set I each player's continuation strategy $s_i^*\big|_I$ maximizes expected payoff given beliefs $\mu_i^*(\cdot \mid I)$ and the other players' continuation strategies $s_{-i}^*\big|_I$.
2. *Belief consistency.* At information sets I reached with positive probability, beliefs $\mu_i^*(\cdot \mid I)$ are obtained from the prior and equilibrium strategies s^* using Bayes' rule.
3. *Separating.* Each provider type chooses a different signal.

Candidate provider strategy profiles are $s_1(H, L) \in \{(C, N), (N, C)\}$.

Profile (C, N)**.** Under belief consistency the operator's beliefs $\mu_2(\cdot \mid m_1)$ are

$$\mu_2(H \mid C) = \frac{P(C \mid H)P(H)}{P(C \mid H)P(H) + P(C \mid L)P(L)} = \frac{1 * 1/3}{1 * 1/3 + 0} = 1, \qquad \mu_2(L \mid C) = 0,$$
$$\mu_2(H \mid N) = 0, \qquad \mu_2(L \mid N) = 1.$$

Under sequential rationality, the operator's strategy is $s_2(C) = D$ and $s_2(N) = R$.

In equilibrium, the provider should not have an incentive to deviate. When $\theta_1 = H$, selecting N decreases provider payoff from $2 \to 0$. When $\theta_1 = L$, selecting C increases provider payoff from $0 \to 1$. So (C, N) is not a separating PBE.

Profile (N, C)**.** Operator beliefs are

$$\mu_2(H \mid C) = 0, \qquad \mu_2(L \mid C) = 1, \qquad \mu_2(H \mid N) = 1, \qquad \mu_2(L \mid N) = 0.$$

The operator's strategy is $s_2(C) = R$ and $s_2(N) = D$. When $\theta_1 = H$, selecting C decreases provider payoff from $3 \to -1$. When $\theta_1 = L$, selecting N increases provider payoff from $-2 \to 3$. So (N, C) is not a separating PBE.

Therefore, there does not exist a separating PBE.

(c) **Pooling PBE.** Now consider a pooling PBE where $s_1(H) = s_1(L) = N$.

The operator's beliefs are

$$\mu_2(H \mid N) = \frac{P(N \mid H)P(H)}{P(N \mid H)P(H) + P(N \mid L)P(L)} = \frac{1 * 1/3}{1 * 1/3 + 1 * 2/3} = 1/3, \quad \mu_2(L \mid N) = 2/3.$$

The operator's strategy is based on expected payoffs

$$E_{\theta_1}[u_2(N, R \mid \theta_1)] = 0, \qquad E_{\theta_1}[u_2(N, D \mid \theta_1)] = 1/3 * 2 + 2/3 * (-2) = -2/3$$

and its strategy is $s_2(N) = R$.

Bayes rule cannot determine $\mu_2(\cdot \mid C)$. Let $\mu_2 := \mu_2(H \mid C)$ and ask what beliefs can discourage both provider types from deviating from N:

$$E_{\theta_1}[u_2(C, R \mid \theta_1)] = 0, \qquad E_{\theta_1}[u_2(C, D \mid \theta_1)] = \mu_2 * 2 + (1 - \mu_2) * (-2) = 4\mu_2 - 2.$$

Comparing the two expected payoffs we see for any $\mu_2 \leq 1/2$ the operator's best response would be $s_2(C) = R$. Choose $\mu_2 = 0$ so $s_2(N) = s_2(C) = R$ and examine the provider's best response.

When $\theta_1 = H$, selecting C decreases payoffs from $0 \to -1$ and when $\theta_1 = L$, selecting C decreases payoff from $0 \to -2$.

Therefore the profile $s^* = (s_1(H) = s_1(L) = N, s_2(C) = s_2(N) = R)$, together with beliefs $\mu_2(H \mid N) = 1/3$ (from Bayes' rule) and $\mu_2(H \mid C) = 0$ (chosen), is a pooling PBE.

(d) **Behavioral strategy for a semi-separating equilibrium.**

Suppose $\sigma_1(C \mid H) = 1,\ \sigma_1(C \mid L) = p \in (0, 1)$.

Operator beliefs.

$$\mu_2(H \mid C) = \frac{\sigma_1(C \mid H)P(H)}{\sigma_1(C \mid H)P(H) + pP(L)} = \frac{1 * 1/3}{1 * 1/3 + 2p/3} = \frac{1}{1 + 2p}, \qquad \mu_2(L \mid C) = \frac{2p}{1 + 2p}$$
$$\mu_2(H \mid N) = 0, \qquad \mu_2(L \mid N) = 1.$$

$\sigma_2(\cdot \mid \mathbf{N})$. When the provider plays N the operator has a pure best response of $s_2(N) = R$ since

$$E_{\theta_1}[u_2(N, D \mid \theta_1)] = 0 * 2 + 1 * (-2) < 0 = E_{\theta_1}[u_2(N, R \mid \theta_1)]$$

So $\sigma_2(R \mid N) = 1 = 1 - \sigma_2(D \mid N)$.

$\sigma_1(\cdot \mid \mathbf{L})$. When the provider plays C, the operator can play a mixed strategy only if both pure strategies are in its best response. This occurs when

$$\begin{aligned} E_{\theta_1 \sim \mu_2}[u_2(C, D \mid \theta_1)] &= 0 = E_{\theta_1 \sim \mu_2}[u_2(C, R \mid \theta_1)] \\ 2 * \mu_2(H \mid C) + -2 * \mu_2(L \mid C) &= 0 \\ 2/(1 + 2p) - 4p/(1 + 2p) &= 0 \\ p &= 1/2 \end{aligned}$$

So $\sigma_1(C \mid L) = 1/2 = \sigma_1(N \mid L)$.

$\sigma_2(\cdot \mid \mathbf{C})$. Let $\sigma_2(D \mid C) = q$ and recall $\sigma_2(R \mid N) = 1$. For a behavioral strategy equilibrium we must find q such that the provider of type $\theta_1 = L$ is indifferent between actions C and N:

$$\begin{aligned} E_{\sigma_2}[u_1(C, \sigma_2 \mid L)] &= E_{\sigma_2}[u_1(N, \sigma_2 \mid L)] \\ q * 1 + (1-q)(-2) &= 0 = u_1(N, R \mid L) \\ q &= 2/3 \end{aligned}$$

So $\sigma_2(D \mid C) = 2/3 = 1 - \sigma_2(R \mid C)$.

$\sigma_1(\cdot \mid \mathbf{H})$ It suffices to show $\sigma_1(C \mid H) = 1 = 1 - \sigma_1(N \mid H)$ is a best response:

$$E_{\sigma_2}[u_1(C, \sigma_2 \mid H)] = 2/3 * 2 + 1/3(-1) = 1 \geq 0 = E_{\sigma_2}[u_1(N, \sigma_2 \mid H)]$$

So $s_1(H) = C$ is a pure strategy best response confirming the check.

Equilibrium. Therefore the following strategy profile

$$\begin{aligned} &\sigma_1(C \mid H) = 1, \quad \sigma_1(N \mid H) = 0, \quad \sigma_1(C \mid L) = 1/2, \quad \sigma_1(N \mid L) = 1/2. \\ &\sigma_2(D \mid C) = 2/3, \quad \sigma_2(R \mid C) = 1/3, \quad \sigma_2(D \mid N) = 0, \quad \sigma_2(R \mid N) = 1. \end{aligned}$$

with beliefs

$$\begin{aligned} \mu_2(H \mid C) &= 1/2, & \mu_2(L \mid C) &= 1/2 \\ \mu_2(H \mid N) &= 0, & \mu_2(L \mid N) &= 1. \end{aligned}$$

is a PBE in behavioral strategies.

Probability of unreliable and deployed.

$$\begin{aligned} P(D, L) &= P(D \mid L)P(L) \\ &= \big[P(D \mid L, C)P(C \mid L) + P(D \mid L, N)P(N \mid L)\big]P(L) && \text{(law of total probability)} \\ &= \big[\sigma_2(D \mid C)\sigma_1(C \mid L) + \sigma_2(D \mid N)\sigma_1(N \mid L)\big]P(L) && \text{(plugging in strategy profiles)} \\ &= \big[\tfrac{2}{3} \cdot \tfrac{1}{2} + 0 \cdot \tfrac{1}{2}\big]\tfrac{2}{3} \\ &= \tfrac{2}{9} \end{aligned}$$

Problem 3 (Traffic routing problem)

(a) **Wardrop equilibrium.** For signal s all paths that carry positive flow must have same expected latency:

$$x^W(\mu) = \begin{cases} 0 & \mu \in [0, \frac{1}{3}] \quad \big(\mu \leq \frac{1}{3} + 2 * 0\big) \\ \frac{\mu - 1/3}{2} & \mu \in [\frac{1}{3}, 1] \\ 1 & \mu \geq \frac{1}{3} + 2 = \frac{7}{3} \ \text{ (not possible)} \end{cases}$$

Conditional expected total latency for $\mu \in [\frac{1}{3}, 1]$ is

$$\begin{aligned} C(\mu) &= \left(1 - \frac{\mu - 1/3}{2}\right)\mu + \left(\frac{\mu - 1/3}{2}\right)\left(\tfrac{1}{3} + 2\frac{\mu - 1/3}{2}\right) \\ &= \mu - \frac{\mu^2 - \mu/3}{2} + \frac{\mu^2 - \mu/3}{2} = \mu \end{aligned}$$

For $\mu \in [0, \frac{1}{3}]$ it's

$$C(\mu) = (1 - 0)\mu + 0(\tfrac{1}{3} + 2 * 0) = \mu$$

So $C(\mu) = \mu$, $\forall \mu \in [0, 1]$.

Uniqueness. Fix μ and suppose the Wardrop equilibrium is not unique. Then there exists $0 \le x < x' \le 1$ such that both flows satisfy the Wardrop condition.
$0 < x' \implies$ path 2 latency is no worse than path 1 latency: $\mu \ge \frac{1}{3} + 2x'$.
$x < 1 \implies$ path 1 latency is no worse than path 2 latency: $\mu \le \frac{1}{3} + 2x$. So

$$\tfrac{1}{3} + 2x' \le \mu \le \tfrac{1}{3} + 2x \implies x' \le \frac{\mu - \frac{1}{3}}{2} \le x$$

which contradicts $x < x'$. Therefore the equilibrium is unique.

(b) **Mean-posterior identity.**

$$\begin{aligned}
\sum_{\{s \in S : \lambda_s > 0\}} \lambda_s \mu_s &= \sum_{\{s \in S : P(s) > 0\}} P(s) P(\theta = 1 \mid s) && \text{(definition)} \\
&= \sum_{\{s \in S : P(s) > 0\}} P(\theta = 1, s) = \sum_{\{s \in S : P(s) > 0\}} P(s \mid \theta = 1) P(\theta = 1) && \text{(Bayes)} \\
&= p \sum_{\{s \in S : P(s) > 0\}} \pi(s \mid \theta = 1) && \text{(law of total probability)} \\
&= p \cdot 1
\end{aligned}$$

Where the final step holds for $P(s) > 0$ since for $p \in (0, 1)$ and every $P(s) = 0$,

$$0 = P(s) = p\pi(s \mid \theta = 1) + (1 - p)\pi(s \mid \theta = 0) \implies \pi(s \mid \theta = 1) = 0$$

(c) **Public signal rule.**

$$\begin{aligned}
C(\mu_s) &= \sum_{\theta} P(\theta \mid s) \cdot C(x^W(\mu_s), \theta) = \mu_s && \text{(conditional expected cost from part (a))} \\
E_s[C(\mu_s)] &= \sum_{s} P(s) C(\mu_s) && \text{(ex ante expected cost)} \\
&= \sum_{s} \lambda_s \mu_s && \text{(from part (a))} \\
&= p && \text{(from part (b))}
\end{aligned}$$

So ex ante expected total latency does not depend on the public signal rule $\pi(s \mid \theta)$ hence any public signal rule gives the same value.

Full revelation. For $s \in \{0, 1\}$, $\pi(s \mid \theta) = 1$ for $s = \theta$ and 0 otherwise. Then

$$\mu(\theta = 1 \mid s) = \frac{\pi(s \mid \theta = 1)p}{\pi(s \mid \theta = 1)p + 0} = 1$$
$$\mu(\theta = 0 \mid s) = \frac{\pi(s \mid \theta = 0)(1 - p)}{\pi(s \mid \theta = 0)(1 - p) + 0} = 1$$

From part (a), $x^W(\mu_s = 1) = (1 - \frac{1}{3})/2 = \frac{1}{3}$.

No information. For $s \in \{0\}$, $\pi(s \mid \theta) = 1$ for $\theta \in \{0, 1\}$. Then

$$\mu(\theta = 1 \mid s) = \frac{\pi(s \mid \theta = 1)p}{\pi(s \mid \theta = 1)p + \pi(s \mid \theta = 0)(1 - p)} = p$$

From part (a),

$$x^W(\mu_s = p) = \begin{cases} 0 & p \in [0, \frac{1}{3}] \\ \frac{p - 1/3}{2} & p \in [\frac{1}{3}, 1] \end{cases}$$

(d) **Private information general rule.** Recommendation probabilities are

$$P(r=2) = P(r=2 \mid \theta=1)P(\theta=1) + P(r=2 \mid \theta=0)P(\theta=0) = q_1 p + q_0(1-p)$$
$$P(r=1) = P(r=1 \mid \theta=1)P(\theta=1) + P(r=1 \mid \theta=0)P(\theta=0) = (1-q_1)p + (1-q_0)(1-p)$$

The posterior probabilities of $\theta = 1$ are

$$P(\theta=1 \mid r=2) = \frac{P(r=2 \mid \theta=1)p}{P(r=2)} = \frac{q_1 p}{q_1 p + q_0(1-p)}$$
$$P(\theta=1 \mid r=1) = \frac{P(r=1 \mid \theta=1)p}{P(r=1)} = \frac{(1-q_1)p}{(1-q_1)p + (1-q_0)(1-p)}$$

Recommendations are incentive compatible for a driver if following the recommendation has expected latency at most as high as deviating. Written below without conditional probability denominators (which cancel as shown in $r=2$)

$$\begin{aligned}
r=2: \qquad & E_{\theta|r=2}[\tfrac{1}{3} + 2q_\theta] \le E_{\theta|r=2}[\theta] \\
& (\tfrac{1}{3} + 2q_1)P(\theta=1 \mid r=2) + (\tfrac{1}{3} + 2q_0)(P(\theta=0 \mid r=2)) \le P(\theta=1 \mid r=2) \\
& \left(\tfrac{1}{3} + 2q_1\right)\frac{q_1 p}{q_1 p + q_0(1-p)} + \left(\tfrac{1}{3} + 2q_0\right)\frac{q_0(1-p)}{q_1 p + q_0(1-p)} \le \frac{q_1 p}{q_1 p + q_0(1-p)} \\
& (\tfrac{1}{3} + 2q_1)q_1 p + (\tfrac{1}{3} + 2q_0)(1-p)q_0 \le q_1 p
\end{aligned}$$

$$\begin{aligned}
r=1: \qquad & E_{\theta|r=1}[\tfrac{1}{3} + 2q_\theta] \ge E_{\theta|r=1}[\theta] \\
& (\tfrac{1}{3} + 2q_1)P(\theta=1 \mid r=1) + (\tfrac{1}{3} + 2q_0)P(\theta=0 \mid r=1) \ge P(\theta=1 \mid r=1) \\
& (\tfrac{1}{3} + 2q_1)(1-q_1)p + (\tfrac{1}{3} + 2q_0)(1-q_0)(1-p) \ge (1-q_1)p
\end{aligned}$$

Bayes rule places $P(r)$ in the denominator of the posterior expectation, so zero-probability recommendations have undefined posterior expectations of $\theta = 1$. These recommendations can safely be dropped from the set of recommendations as are all $s : \lambda_s = 0$ in part (c) since for $p \in (0,1)$ and $P(r) = 0$

$$0 = P(r) = pP(r \mid \theta=1) + (1-p)P(r \mid \theta=0) \implies P(r \mid \theta=1) = 0.$$

Expected platform costs is

$$\begin{aligned}
E_\theta\left[C(q_\theta, \theta)\right] = J(q_0, q_1) &= (1-p)(C(q_0, 0)) + p(C(q_1, 1)) \\
&= (1-p)\left[(1-q_0)0 + q_0\left(\tfrac{1}{3} + 2q_0\right)\right] + p\left[(1-q_1)1 + q_1\left(\tfrac{1}{3} + 2q_1\right)\right] \\
&= (1-p)\left[2q_0^2 + \tfrac{1}{3}q_0\right] + p\left[2q_1^2 - \tfrac{2}{3}q_1 + 1\right]
\end{aligned}$$

(e) **Socially optimal flow, costs, and IC compatibility.**

Socially optimal flow.

$\theta = 1$: $\quad C(x,1) = 1 - x + \frac{x}{3} + 2x^2 = 2x^2 - \frac{2x}{3} + 1$

$\frac{dC(x,1)}{dx} = 0 \implies 0 = 4x - \frac{2}{3} \implies x = \frac{1}{6}$ (This is a minimum since second derivative is positive)

$\theta = 0$: $\quad C(x,0) = \frac{x}{3} + 2x^2$

$\frac{dC(x,0)}{dx} = 0 \implies 0 = \frac{1}{3} + 4x \implies x = -\frac{1}{12}$ (no critical point on $(0,1)$ so minimum is on the boundary)

$C(0,0) = 0, \quad C(1,0) = \frac{1}{3} + 2 = \frac{7}{3}$ (so x=0 at the boundary achieves minimum)

So the optima are $(x_0^S, x_1^S) = (0, 1/6)$.

Ex ante expected cost.

$$E_\theta[C(x,\theta)] = p \cdot (\tfrac{5}{6} + \tfrac{1}{6}(\tfrac{1}{3} + \tfrac{2}{6})) + (1-p)0 = p(\tfrac{15}{18} + \tfrac{2}{18}) = \tfrac{17}{18}p$$

Private recommendation rule. Plugging in socially optimal flows $(q_0, q_1) = (0, 1/6)$ into the inequalities from part (d)

$$r = 2: \quad (1/3 + 1/3)(p/6) + (1/3 + 2 \cdot 0)(1-p) \cdot 0 \le p/6 \implies 2p/18 \le p/6 \implies 2p \le 3p$$
($r = 2$ is incentive compatible $\forall\, p \in (0,1)$)

$$r = 1: \quad (1/3 + 1/3)(5p/6) + (1/3 + 0)(1-p) \ge 5p/6 \implies 11p/18 \le 1/3 \implies p \le 6/11$$

so recommendations r are incentive compatible for $p \le 6/11$, determined by $r = 1$.

Compare to public information rules. We've just shown that for $p \le 6/11$, a private information rule can achieve the ex ante expected latency at $17p/18$, which for $p \in (0,1)$ is lower than the public rule ex ante expected latency from part (c) of p.

A private rule can give different drivers different posteriors, depending on which recommendation they receive. Knowing the (q_0, q_1) that achieve the social optimum, we see posteriors are

$$P(\theta = 1 \mid r = 2) = \tfrac{p}{6}/(\tfrac{p}{6} + 0) = 1,$$
$$P(\theta = 1 \mid r = 1) = \tfrac{5p}{6}/(\tfrac{5p}{6} + (1-p)).$$

Drivers told $r = 2$ know with certainty $\theta = 1$ and follow the recommendation. Drivers told $r = 1$ remain uncertain enough that the recommendation path looks no worse than the deviation. So with these posteriors on the recommendations, the planner can hold the socially optimal flows.

Under any public rule all drivers hold the same μ_s, the cost is $C(\mu_s) = \mu_s$, and the ex ante average is p by part (b).

(f) **Cost minimization.**

$$\min_{q_0,q_1} \quad (1-p)\left[2q_0^2 + \tfrac{1}{3}q_0\right] + p\left[2q_1^2 - \tfrac{2}{3}q_1 + 1\right] \qquad (J(q_0,q_1))$$
$$\text{s.t.} \quad (\tfrac{1}{3} + 2q_1)q_1 p + (\tfrac{1}{3} + 2q_0)(1-p)q_0 \le q_1 p \qquad (r=2 \text{ incentive compatibility})$$
$$(\tfrac{1}{3} + 2q_1)(1-q_1)p + (\tfrac{1}{3} + 2q_0)(1-q_0)(1-p) \ge (1-q_1)p \qquad (r=1 \text{ incentive compatibility})$$
$$(q_0, q_1) \in [0,1]^2$$

```
# %%
import numpy as np
import cvxpy as cp
import polars as pl

pl.Config.set_tbl_cols(-1)

# %%
# ======================
# Problem 3 Part (f)
# ======================
def private_recs(p):
    # variables
    q0 = cp.Variable()
    q1 = cp.Variable()
    # constraints
    const = [
        q0 >= 0,
        q0 <= 1,
        q1 >= 0,
```

```
        q1 <= 1,
    ]
    const += [
        (q1 / 3 + 2 * cp.square(q1)) * p + (q0 / 3 + 2 * cp.square(q0)) * (1 - p)
        <= q1 * p
    ]
    const += [
        (1 / 3 + 5 * q1 / 3 - 2 * cp.square(q1)) * p
        + (1 / 3 + 5 * q0 / 3 - 2 * cp.square(q0)) * (1 - p)
        >= (1 - q1) * p,
    ]
    # objective
    obj = (1 - p) * (2 * cp.square(q0) + q0 / 3) + p * (
        2 * cp.square(q1) - 2 * q1 / 3 + 1
    )
    # problem
    prob = cp.Problem(cp.Minimize(obj), const)
    prob.solve()
    return float(q0.value), float(q1.value), float(prob.value), prob.status


results = []
for p in np.arange(1, 11):
    q0, q1, obj, status = private_recs(p / 11)
    public_cost = p / 11
    soc_opt_cost = (p / 11) * (17 / 18)
    public_gain = public_cost - obj
    soc_opt_gap = obj - soc_opt_cost
    results.append(
        (
            f"{p}/11",
            round(q0, 4),
            round(q1, 4),
            round(obj, 4),
            round(public_cost, 4),
            round(public_gain, 4),
            round(soc_opt_cost, 4),
            round(soc_opt_gap, 4),
        )
    )
```

p	rule_q0	rule_q1	expected_cost	public_cost	public_gain	soc_opt_cost	soc_opt_gap
1/11	0.0000	0.1667	0.0859	0.0909	0.0051	0.0859	0.0000
2/11	0.0000	0.1667	0.1717	0.1818	0.0101	0.1717	0.0000
3/11	0.0000	0.1667	0.2576	0.2727	0.0152	0.2576	0.0000
4/11	0.0000	0.1667	0.3434	0.3636	0.0202	0.3434	0.0000
5/11	0.0000	0.1667	0.4293	0.4545	0.0253	0.4293	0.0000
6/11	0.0000	0.1667	0.5152	0.5455	0.0303	0.5152	0.0000
7/11	0.0000	0.2124	0.6037	0.6364	0.0327	0.6010	0.0027
8/11	0.0000	0.2500	0.6970	0.7273	0.0303	0.6869	0.0101
9/11	0.0258	0.2758	0.7940	0.8182	0.0242	0.7727	0.0213
10/11	0.0534	0.3034	0.8947	0.9091	0.0144	0.8586	0.0361

Problem 4 (Employment recommendations)

(a) **Payoff maximizing company actions.** Let $\mu := P(\omega = G \mid i) \in [0,1]$ be the posterior belief of G given information i. Expected payoffs under the posterior are

$$E_\mu[u(F,\mu)] = 2\mu - 4(1-\mu) = 6\mu - 4, \quad E_\mu[u(I,\mu)] = \mu - (1-\mu) = 2\mu - 1, \quad E_\mu[u(R,\mu)] = 0.$$

Then payoff-maximizing actions a^* are

$$a^* = \begin{cases} F & 6\mu - 4 \geq 2\mu - 1 \implies \mu \in [3/4, 1] \\ I & 2\mu - 1 \geq 0 \implies \mu \in [1/2, 3/4] \\ R & \mu \in [0, 1/2] \end{cases}$$

Where at the seams $\{1/2, 3/4\}$ either action is maximizing.
Under full revelation, the company's payoff-maximizing actions are

$$a^{FR*} = F \text{ if } \omega = G \text{ else } R$$

So the school's expected payoff is

$$E_\omega[v(a^{FR*}, \omega)] = (2/3) \cdot 2 + (1/3) \cdot 0 = 4/3$$

Under no information, the company's expected payoffs are

$$E_\omega[u(F,\omega)] = 2 \cdot (2/3) - 4 \cdot (1/3) = 0, \quad E_\omega[u(I,\omega)] = (2/3) - (1/3) = 1/3, \quad E_\omega[u(R,\omega)] = 0,$$

So the company's payoff maximizing action under no information is $a^{NI*} = I$ and the school's expected payoffs are $E_\omega[v(I,\omega)] = 3/2$, which is higher than the full-revelation expected payoffs.

(b) **Recommendation.** The recommendations' probabilities are

$$P(I) = P(I \mid G)P(G) + P(I \mid M)P(M) = 2x/3 + y/3, \quad P(F) = 1 - P(I) = 1 - 2x/3 - y/3.$$

Posterior beliefs are

$$\mu_I := P(G \mid I) = \frac{P(I \mid G)P(G)}{P(I)} = \frac{2x/3}{2x/3 + y/3} = \frac{2x}{2x + y},$$
$$\mu_F := P(G \mid F) = \frac{P(F \mid G)P(G)}{P(F)} = \frac{2(1-x)/3}{2(1-x)/3 + (1-y)/3} = \frac{2 - 2x}{3 - 2x - y},$$

and $\mu_r(M \mid r) = 1 - \mu_r$ for $r \in \{I, F\}$.
Incentive compatibility conditions require following the recommendation to yield expected payoffs at least as good as any deviation:

$r = I$:

$$\begin{aligned} E_{\omega\sim\mu_I}[u(I,\omega)] &\geq E_{\omega\sim\mu_I}[u(F,\omega)] \\ \mu_I u(I,G) + (1-\mu_I)u(I,M) &\geq \mu_I u(F,G) + (1-\mu_I)u(F,M) \\ \mu_I - 1 + \mu_I &\geq 2\mu_I - 4 + 4\mu_I \implies \mu_I \leq 3/4 \\ 2x &\leq 3x/2 + 3y/4 \implies x \leq 3y/2 \end{aligned}$$

$$\begin{aligned} E_{\omega\sim\mu_I}[u(I,\omega)] &\geq E_{\omega\sim\mu_I}[u(R,\omega)] \\ \mu_I - 1 + \mu_I &\geq 0 \implies \mu_I \geq 1/2 \\ 2x &\geq x + y/2 \implies x \geq y/2 \end{aligned}$$

$r = F$:

$$\begin{aligned} E_{\omega\sim\mu_F}[u(F,\omega)] &\geq E_{\omega\sim\mu_F}[u(I,\omega)] \\ 2\mu_F - 4 + 4\mu_F &\geq \mu_F - 1 + \mu_F \implies \mu_F \geq 3/4 \\ 2 - 2x &\geq 9/4 - 3x/2 - 3y/4 \implies x \leq 3y/2 - 1/2 \end{aligned}$$

$$\begin{aligned} E_{\omega\sim\mu_F}[u(F,\omega)] &\geq E_{\omega\sim\mu_F}[u(R,\omega)] \\ 2\mu_F - 4 + 4\mu_F &\geq 0 \implies \mu_F \geq 2/3 \\ 2 - 2x &\geq 2 - 4x/3 - 2y/3 \implies x \leq y \end{aligned}$$

Note μ_I, μ_F are only defined through Bayes rule when $P(I), P(F)$ are positive. When $r : P(r) = 0$, we have $0 = 2P(r \mid G)/3 + P(r \mid M)/3 \implies P(r \mid \omega) = 0$, so the conditional recommendations also have zero probability and can be safely dropped.

(c) **School's optimization.** Maximize recommendation payoffs, constrained by incentive compatibility:

$$\begin{aligned}
\max_{x,y} \quad & E_{\omega,x,y}[v(r(x,y),\omega)] \\
\text{s.t.} \quad & E_{\omega\sim\mu_I}[u(I,\omega)] \geq E_{\omega\sim\mu_I}[u(F,\omega)], \qquad E_{\omega\sim\mu_I}[u(I,\omega)] \geq E_{\omega\sim\mu_I}[u(R,\omega)], \\
& E_{\omega\sim\mu_F}[u(F,\omega)] \geq E_{\omega\sim\mu_F}[u(I,\omega)], \qquad E_{\omega\sim\mu_F}[u(F,\omega)] \geq E_{\omega\sim\mu_F}[u(R,\omega)], \\
& (x,y) \in [0,1]^2.
\end{aligned}$$

Written in terms of x, y from part (b) this problem is

$$\begin{aligned}
\max_{x,y} \quad & \tfrac{2}{3}(3x/2 + 2\cdot(1-x)) + \tfrac{1}{3}(3y/2 + 2\cdot(1-y)) = 2 - x/3 - y/6 \\
\text{s.t.} \quad & x \geq y/2, \qquad x \leq 3y/2 - 1/2, \qquad (x,y) \in [0,1]^2,
\end{aligned}$$

where the constraints are simplified since $x \leq 3y/2 - 1/2$ implies $x \leq y$ and $x \leq 3y/2$ on $(x,y) \in [0,1]^2$.

The codeblock below computes the school's optimal rule $(x,y) = (1/4, 1/2)$ and optimal expected payoff of $1.833 = 11/6$.

```
# ======================
# Problem 4 Part (c)
# ======================
x = cp.Variable()
y = cp.Variable()
const = [
    x >= 0,
    x <= 1,
    y >= 0,
    y <= 1,
]
const += [x >= y / 2]
const += [x <= 3 * y / 2 - 1 / 2]
obj = 2 - x / 3 - y / 6
prob = cp.Problem(cp.Maximize(obj), const)
prob.solve()
print("x:\t\t", np.round(x.value, 4))
print("y:\t\t", np.round(y.value, 4))
print("optimal payoff:\t", np.round(obj.value, 4))

# rule_x: 0.25
# rule_y: 0.5
# optimal payoff: 1.8333
```

(d) **Optimal rule posterior beliefs.** At $(x,y) = (1/4, 1/2)$, posterior beliefs are

$$\mu_I = \frac{2x}{2x+y} = \frac{1/2}{1/2+1/2} = 1/2, \qquad \mu_F = \frac{2-2x}{3-2x-y} = \frac{2-1/2}{3-1/2-1/2} = 3/4.$$

The company's ex ante payoff under this rule is

$$\begin{aligned}
E_r[E_{\omega\sim\mu_r}[u(r,\omega)]] &= P(I)\big[\mu_I u(I,G) + (1-\mu_I)u(I,M)\big] + \big(1-P(I)\big)\big[\mu_F u(F,G) + (1-\mu_F)u(F,M)\big] \\
&= (2x/3 + y/3)(4x/(2x+y) - 1) + (1 - 2x/3 - y/3)((12-12x)/(3-2x-y) - 4) \\
&= 1/3\cdot 0 + 2/3\cdot 1/2 = 1/3.
\end{aligned}$$

From part (a), the company's payoff maximizing action under no information is $a^{NI*} = I$ and its ex ante expected payoff is $E_\omega[u(I,\omega)] = 1/3$.

Also from part (a), the company's payoff maximizing action under full revelation is $a^{FR*} = F$ if $\omega = G$ else R and its ex ante expected payoff is $E_\omega[u(a^{FR*},\omega)] = 2/3\cdot 2 + 0 = 4/3$.

The company's preferred policy is full information, where its expected payoff is $4/3$. The school's preferred policy is the messaging policy, where its expected payoff is $11/6$. These need not coincide because the school's payoffs are independent of the state ω so it has a different optimal policy that performs best under partial information, while the company's payoffs are a direct mapping to state ω and it does best under full information.